

MeetSync
A PROJECT REPORT
for
Major Project-1 (CA401P)
Session (2025-26)

Submitted by
Prince Kumar
(202410116100149)
Priyanshi
(202410116100153)
Priyanshu Kansal
(202410116100155)
Rahul
(202410116100157)

Submitted in partial fulfilment of the
Requirements for the Degree of

MASTER OF COMPUTER APPLICATION

Under the Supervision of
Ms. Shweta Singh
Assistant Professor



Submitted to
Department Of computer Applications
Kiet Group of Institutions, Ghaziabad,
Uttar Pradesh -201206
(June-2026)

CERTIFICATE

Certified that **Prince Kumar 202410116100149, Priyanshi 202410116100153, Priyanshu Kansal 202410116100155 and Rahul 202410116100157** has/have carried out the project work having “**MeetSync**” (**Major Project-CA301P**) for **Master of Computer Application** from Dr. A.P.J. Abdul Kalam Technical University (AKTU) (formerly UPTU), Lucknow under my supervision. The project report embodies original work, and studies are carried out by the student himself/herself and the contents of the project report do not form the basis for the award of any other degree to the candidate or to anybody else from this or any other University/Institution.

Ms. Shweta Singh
Assistant Professor
Department of Computer Applications
KIET Group of Institutions, Ghaziabad

Dr. Sachin Malhotra
Dean
Department of Computer Applications
KIET Group of Institutions, Ghaziabad

ABSTRACT

The project titled “**MeetSync**” is an intelligent and innovative solution developed to enhance the way individuals receive career guidance and professional development support. Traditional career counseling methods often depend on manual assessment, limited availability of experts, and generic advice that may not align with individual aspirations. To address these challenges, this project introduces an AI-driven digital platform that delivers personalized, accurate, and accessible career guidance, ensuring clarity and confidence in career decision-making.

The system is designed using artificial intelligence techniques such as machine learning and natural language processing to analyze user data including skills, interests, educational background, and career objectives. Based on this analysis, the MeetSync recommends suitable career paths, required skill sets, learning resources, and potential job roles. The platform offers an interactive and user-friendly interface that enables users to receive real-time insights and actionable recommendations tailored to their individual profiles.

By bridging the gap between academic knowledge and industry requirements, the application helps users identify skill gaps and suggests relevant courses, certifications, and improvement strategies. This approach not only improves employability but also supports long-term career planning. The system adapts to evolving job market trends, ensuring that recommendations remain relevant and up to date.

The proposed application also provides scope for future enhancements such as resume analysis, interview preparation modules, integration with job portals, and AI-based career progress tracking. Overall, the MeetSync demonstrates the effective use of artificial intelligence in delivering scalable, efficient, and personalized career guidance.

Keywords: Artificial Intelligence, Career Guidance System, Personalized Recommendations, Skill Gap Analysis, Career Development

ACKNOWLEDGEMENTS

Success in life is never achieved alone. I express my deepest gratitude to my thesis supervisor, Dr. Amit Kumar (Associate Professor), for her guidance, assistance, and encouragement throughout my project work. Her insightful ideas, comments, and suggestions have been invaluable in helping me complete this project successfully.

I am also immensely grateful to Dr. Sachin Malhotra, Professor and Head of, the Department of Computer Applications, for his valuable feedback and administrative support on several occasions. I am fortunate to have many supportive friends who have been of great help during critical moments of this journey.

Lastly, I would like to extend my heartfelt thanks to my family members and all those who have provided me with moral support and assistance, both directly and indirectly. Completing this work on time would not have been possible without their unwavering support. Their constant love and encouragement have filled my life with joy and happiness.

Prince Kumar

Priyanshi

Priyanshu Kansal

Rahul

TABLE OF CONTENTS

Certificate.....	ii
Abstract.....	iii
Acknowledgment.....	iv
Table of Contents	v–vi
1. Introduction	7–10
1.1 Background.....	7
1.2 Problem Review	7
1.3 Objective.....	8
1.4 Key Features	9
1.5 Scope	9
1.6 Hardware/Software Requirements.....	10
1.7 Background Study	10
2. Feasibility Study 11	13
2.1 Economical Feasibility	11
2.2 Technical Feasibility.....	12
2.3 Operational Feasibility.....	12
2.4 Behavioral Feasibility.....	13
3. Software Requirement Specification 14.....	20
3.1 Functionalities 14.....	16
3.2 User and Characteristics 16.....	17
3.3 Features of Project 17	18
3.4 Features of Admin 18	19
3.5 Features of User 19.....	20
4. Hardware and Software Requirements	21
5. Project Flow 22.....	23
6. System Requirement	24–28
6.1 Functional Requirements	24–25
6.2 Non-Functional Requirements	26–27
6.3 Design Goals.....	27–28

7. System Design.....	29–34
7.1 Primary Design Phase	29–30
7.2 Secondary Design Phase	31–32
7.3 User Interface Design	33–34
8. Project Codes.....	35–37
9. Project Outcomes	38–46
9.1 User Module	38-39
9.2 Admin Module	39-42
9.3 Database Module.....	43–46
10. Testing of MeetSync Project	46-51
10.1 Introduction to Testing.....	46
10.2 Types of Testing	47-48
10.3 Test Plan and Execution.....	49-50
10.4 Results and Reports (15 test cases).....	50-51
11. Proposed Time Duration	52-53
Conclusion.....	54
References	55-56

CHAPTER - 1

INTRODUCTION

11.1 Background

Career guidance plays a crucial role in shaping an individual's professional life. Traditionally, career counseling has been provided through human experts, aptitude tests, and academic advisors. However, these methods are often time-consuming, costly, and limited in personalization. With the rapid growth of technology and the availability of large-scale data, artificial intelligence has emerged as a powerful tool to address these challenges.

The **MeetSync** project is developed to leverage artificial intelligence and modern web technologies to provide personalized, scalable, and accessible career guidance. The system analyzes user skills, interests, and goals to suggest suitable career paths and learning roadmaps.

11.2 Problem Review

Many students and professionals face confusion while choosing a career due to lack of proper guidance, outdated information, and limited access to experts. Traditional systems provide generalized advice that does not adapt to individual needs.

Some major problems include:

- Lack of personalized career recommendations
- Limited awareness of industry skill requirements
- Difficulty in tracking career progress
- High dependency on manual counseling

The MeetSync aims to overcome these issues by providing automated, data-driven, and user-centric career guidance.

11.3 Objective

The main objectives of the MeetSync project are:

- To provide personalized career recommendations using AI
- To analyze user skills, interests, and goals
- To suggest relevant skills, courses, and learning paths
- To make career guidance accessible and affordable
- To ensure secure storage of user data
- To reduce dependency on traditional counseling methods

11.4 Key Features

- AI-based career recommendations
- User authentication using Email and Google OAuth
- Skill gap analysis and learning roadmap
- Interactive user dashboard
- Secure data storage using PostgreSQL
- Responsive and user-friendly interface

11.5 Scope

The scope of the project includes students, fresh graduates, and working professionals seeking career guidance. The system focuses on career recommendation, skill development suggestions, and progress tracking.

Future scope includes:

- Resume analysis
- Interview preparation modules
- Job portal integration

11.6 Hardware & Software Requirements

Hardware

11.6.1 Laptop/PC

11.6.2 Minimum 4GB RAM

Software

11.6.3 VS Code

11.6.4 Node.js

11.6.5 Express.js

11.6.6 React.js

11.6.7 MongoDB Atlas

11.6.8 Chrome / Firefox Browser

11.6.9 GitHub (for version control)

11.7 Background

Earlier career guidance systems relied heavily on expert knowledge and static assessments. Recent advancements in AI, machine learning, and recommender systems have enabled dynamic, personalized career coaching platforms. Applications like LinkedIn and Coursera demonstrate the effectiveness of AI-driven career insights, which inspired the development of this project.

CHAPTER - 2

FEASIBILITY STUDY

A feasibility study is a critical phase in software development that evaluates the practicality and viability of a proposed system before full-scale implementation. It helps determine whether the project is technically sound, economically viable, operationally acceptable, and socially adaptable. For the **MeetSync** project, a detailed feasibility analysis was conducted to ensure successful development, deployment, and adoption of the system.

2.1 Economic Feasibility

Economic feasibility assesses whether the project can be developed and maintained within reasonable cost limits while delivering value to its users. The MeetSync is economically feasible as it primarily utilizes open-source technologies such as **React.js**, **Node.js**, **PostgreSQL**, and **Tailwind CSS**, which significantly reduce software licensing costs.

The development cost is minimal since no proprietary tools or paid APIs are mandatory for core functionality. Deployment on cloud platforms such as **Vercel** further reduces infrastructure expenses, as it offers free or low-cost hosting options for academic and small-scale projects. Additionally, automation of career guidance minimizes the need for human counselors, leading to long-term cost savings.

2.2 Technical Feasibility

Technical feasibility examines whether the proposed system can be developed using the available technology, tools, and expertise. The MeetSync is technically feasible due to the availability of mature and widely adopted technologies in the current software ecosystem.

The project is built using a modern full-stack architecture, with **React.js** for frontend development, **Node.js and Express.js** for backend services, and **PostgreSQL** for secure data storage. Artificial intelligence concepts such as recommendation systems and data analysis are implemented using established algorithms and APIs. The development team possesses adequate technical knowledge in web development, database management, and basic AI concepts, ensuring smooth implementation. Furthermore, the modular architecture of the system allows easy debugging, maintenance, and future enhancements, confirming the technical viability of the project.

Furthermore, the modular architecture of the system allows easy debugging, maintenance, and future enhancements, confirming the technical viability of the project.

2.3 Operational Feasibility

Operational feasibility evaluates how effectively the system can function in a real-world environment and whether users can operate it comfortably. The MeetSync is designed with a simple, intuitive, and user-friendly interface that requires minimal training.

Users can easily register, create profiles, and receive career recommendations through a

guided workflow. The system automates complex processes such as skill analysis and recommendation generation, reducing manual effort for both users and administrators. From an administrative perspective, the system allows efficient monitoring and management of user data and system performance.

Due to its ease of use, automation, and clear user interaction flow, the system is highly operationally feasible and suitable for real-world deployment.

2.4 Behavioral Feasibility

Behavioral feasibility focuses on user acceptance and adaptability of the system. The success of any software application depends largely on how well it is received by its intended users. The MeetSync is designed to be user-centric, ensuring transparency, fairness, and data privacy.

Since users receive personalized recommendations based on their own inputs, they are more likely to trust and engage with the system. The platform avoids biased suggestions by focusing on skill-based analysis rather than demographic factors. Secure authentication and data protection measures enhance user confidence and acceptance.

As career guidance is a widely recognized need among students and professionals, the system is expected to be positively received, making it behaviorally feasible.

CHAPTER - 3

SOFTWARE REQUIREMENTS SPECIFICATION

The Software Requirements Specification (SRS) describes all functional, non-functional, user-based, and system-level requirements necessary for the successful development of the **MeetSync System**. The purpose of this chapter is to ensure that every requirement is clearly defined, validated, and understood before the development and testing phases. The SRS acts as a communication bridge between developers, users, administrators, and other stakeholders involved in the project.

The core functionalities of the MeetSync System are as follows:

1. User Registration & Authentication

- New users can create an account using name, email, phone, and password.
- Password is stored securely using hashing/encryption.
- Admin has a separate login panel.

2. User Profile Management

- Users can create and update their profile details such as:
 - Educational background
 - Skills
 - Interest
- Profile data is stored securely in the database.

3. AI-Based Career Recommendation

- The system analyzes user profile data using AI algorithms.
 - Skills
 - Interest
 - Education
 - Market Demand
 - Users receive personalized career suggestions.

4. Skill Gap Analysis

- The system identifies missing or weak skills for selected career paths.
 - Beginner
 - Intermediate
 - Advanced
- System automatically checks availability based on selected dates.
- Total amount calculation = price per day × number of rental days.
- Booking is auto-confirmed once submitted.
- Booking details stored in MongoDB Atlas.

5. Learning Resource Recommendation

The system recommends courses, certifications, and learning materials.

- Resources are mapped to required skills.
- Users can save recommended resources for future reference.

- Users can also cancel a booking before pickup date.

6. Admin Dashboard

Admin has complete control of the system, including:

- View total users and activity statistics.
- Manage career categories and skill data.
- Monitor system usage and performance.
- Ensure data integrity and security.

3.1 User and Characteristics

1. End Users (Students / Professionals)

These users seek guidance for career planning and skill development.

Characteristics:

- Basic knowledge of online booking. Basic computer and internet knowledge.
- Access to mobile or desktop devices.
- Interested in career growth and learning.
- Expect simple, fast, and personalized recommendations.

2. Administrator

Responsibilities:

- Manage career data and recommendations.
- Monitor user activity.

- Users
- Reports

Characteristics:

- Must have login credentials.
- Intermediate computer knowledge.
- Responsible for keeping data updated.

3. System Requirements for Users

- A modern browser (Chrome/Firefox).
- Stable internet connection.
- Ability to view images and dynamic content.

3.2 Features of Project

This project provides the following major features:

1. Personalized Career Guidance

AI-driven recommendations tailored to individual users.

2. Skill-Based Analysis

Career suggestions based on skill evaluation and interests.

3. Secure Login System

- Password hashing
- Role-based authentication
- Error messages for wrong credentials

4. View Learning Roadmap

Step-by-step guidance from beginner to advanced levels.

- Specifications

- Pricing
- Photos
- Terms & conditions

5. Transparent Booking

Users can view:

- Booking date
- Car details
- Amount paid
- Booking status

6. Easy Booking Cancellation

Users can cancel bookings instantly, and the cancellation status updates in database.

7. Admin Reporting

Admin gets summary of:

- Total cars
- Total bookings
- Revenue insights (if extended)
- Booking history

8. Cloud Database Integration

All data is stored in **MongoDB Atlas**, ensuring:

- Scalability
- Remote access
- Data security

3.3 Features of Admin

Admin has full authority and advanced controls such as:

1. User Management

- View registered users.
- Monitor user activity.
- Manage user access if required.

2. Career & Skill Management

- Add or update career options
- Modify skill requirements.
- Maintain learning resources.

3. Dashboard Monitoring

Admin dashboard shows:

- Total vehicles in inventory
- Total confirmed bookings
- Today's bookings
- Monthly analytics

4. User Management (Optional Extension)

- View user details
- Identify frequent customers
- Blacklist users in case of misuse

3.4 Features of User

Users enjoy the following functionalities:

1. Account Creation & Login

Allows secure login and profile creation.

2. Profile Setup

Users can add and update skills, education, and interests.

3. Career Recommendations

Users can search based on:

- Price
- Category
- Seating capacity

4. Skill Roadmap

Users can book cars easily by selecting dates and confirming details.

5. Save Recommendation

Users can bookmark careers and resources.

CHAPTER - 4

HARDWARE AND SOFTWARE REQUIREMENTS

Hardware Requirements

1. **Processor:** Intel Core i5 or higher (Quad-Core recommended)
2. **RAM:** Minimum 8 GB (16 GB recommended for smoother development)
3. **Storage:** 500 GB HDD / 256 GB SSD or higher
4. **Display:** 15-inch monitor with minimum resolution of 1366×768
5. **Network:** Stable internet connection for hosting and deployment

Software Requirements

1. **Operating System:** Windows / Linux / MacOS
2. **Frontend Technology:** React.js for building dynamic and responsive user interfaces
3. **Backend Technology:** Node.js with Express.js for server-side development
4. **Database:** MongoDB for flexible, NoSQL-based data storage
5. **Web Browser:** Google Chrome / Mozilla Firefox (latest version)
6. **Code Editor / IDE:** Visual Studio Code (VS Code)
7. **Version Control:** Git and GitHub for collaboration and source code management

CHAPTER - 5

PROJECT FLOW

The development of the **MeetSync** follows a structured **Software Development Life Cycle (SDLC)** approach. The key phases are:

1. Analysis & Planning

- Identify the problem statement and limitations of existing car rental systems.
- Gather requirements from both end-users (customers) and rental agencies.
- Define the project scope, features, and technology stack.
- Prepare a development plan with timelines and responsibilities.

2. Frontend Development

- Implement the designed UI using **React.js**.
- Build reusable components for login, car browsing, booking, and payment.
- Ensure responsiveness across multiple devices (desktop, tablet, mobile).

3. Backend Development

- Set up server-side operations using **Node.js with Express.js**.
- Create RESTful APIs to handle booking requests, user authentication, and payments.
- Implement error handling and ensure data security.

4. Database Integration

- Design and integrate **MongoDB** collections for storing user details, car availability, and booking history.

- Ensure proper indexing and relationships for faster data retrieval.
- Enable real-time updates for car availability status.

5. Testing & Debugging

- Conduct unit testing for individual modules and integration testing for workflows.
- Perform functional, usability, and security testing.
- Debug errors and refine the system for stability and reliability.

6. Maintenance & Updates

- Monitor system performance post-deployment.
- Fix bugs, optimize performance, and roll out updates.
- Add future enhancements like **AI-based recommendations, and loyalty programs** as per roadmap.

CHAPTER – 6

SYSTEM REQUIREMENT

System requirements define the essential conditions, functionalities, constraints, and quality standards needed for the successful development and execution of the Car Rental Booking Application. This chapter outlines both **functional** and **non-functional** requirements and the design goals that guide the structural and technical planning of the system.

6.1 Functional Requirements

Functional requirements describe **what the system should do**. These are directly linked to the core operations of the Car Rental Booking Application.

1. User Registration & Authentication

- The system must allow new users to create an account.
- The user must provide valid email, phone number, name, and secure password.
- Login should validate registered credentials and allow access to the dashboard.
- Passwords must be stored in a hashed format.

2. User Login & Dashboard

- Users must be able to log in securely.
- After login, users should see their dashboard displaying active and past bookings.

3. Career Recommendation System

- The system shall analyze user profile data using AI-based logic.
- The system shall generate personalized career recommendations.
- The system shall display recommended career paths clearly on the user dashboard.

4. Skill Gap Analysis

- The system shall identify skill gaps based on selected career paths.
- The system shall categorize skills into beginner, intermediate, and advanced levels.
- The system shall provide a learning roadmap to bridge skill gaps.

5. Learning Resource Recommendation

- The system shall recommend courses, certifications, and learning materials.
- Overlapping dates must be blocked automatically.
- Availability The system shall associate learning resources with specific skills.

6. Dashboard and Progress Tracking

- The system shall provide a personalized dashboard for each user.
- The system shall display saved career paths and skill roadmaps.
- Booking must be stored in MongoDB Atlas.
- Booking should automatically get the status “**Confirmed**”.

7. Admin Management

- The system shall provide a secure admin login..
- Database must update booking status immediately.
- UI should reflect updated status on refresh.

8. Admin Module

Admin must be able to:

- **Add Users**
Upload details such as name.
- **Edit Details**
Update any user information.
- **Delete Users**
Remove unactive users.
- **Admin Dashboard**
Summary of:
 - Total Users
 - Total Registered Students
- CRUD operations should be efficient and validated.

6.2 Non-Functional Requirements

These define **how the system should behave**, focusing on performance, security, and user experience.

1. Performance Requirements

- System should respond to user actions within 1–3 seconds.
- Searching, filtering, and booking must load without lag.
- Backend APIs must be optimized to handle multiple users.

2. Security Requirements

- Passwords must be hashed (e.g., bcrypt).
- Only admin should access admin panel.
- All input fields must be validated.
- Secure communication between frontend and backend.

3. Usability Requirements

- Clean UI with simple navigation.
- Minimal steps for booking a car.
- Buttons, forms, and pages must be clearly labeled.

4. Reliability Requirements

- System should work consistently without failure.
- Booking system must always ensure correct availability checking.
- Data retrieved from MongoDB must always be accurate.

5. Availability Requirements

- System should remain accessible 24/7.
- Cloud database ensures high uptime availability.

6. Scalability Requirements

- The system should support:
 - More users
 - More cars
 - More booking records
- MongoDB Atlas supports easy scaling.

7. Portability Requirements

- Should work on all modern browsers:

- Chrome
- Firefox
- Edge
- Brave
- UI should adapt to desktops, laptops, tablets, and mobiles.

8. Maintainability Requirements

- Code must be modular with separate components for UI.
- Database schemas must allow new fields without major changes.

9. Compatibility Requirements

- Frontend should be compatible across devices.
- Backend API must be compatible with any frontend framework if expanded in future.

10. Responsiveness Requirements

- UI must adjust based on screen width.
- Pages must remain readable even on mobile devices.
- Navigation menus must change layout for smaller screens.

6.3 Design Goals

1. Performance Design Goal

- Ensure fast data loading from MongoDB.
- Optimize API endpoints.
- Efficient state management using React hooks.

2. Security Design Goal

- Enforce strong authentication.
- Validate all inputs on frontend and backend.

3. UI/UX Design Goal

- Provide visually simple, modern, and responsive interface.
- Avoid clutter and maintain consistent theme.

4. Maintainability Design Goal

- Modular React components.
- Clean Express route structure.
- Reusable backend controllers.
- Centralized error handling.

CHAPTER – 7

SYSTEM DESIGN

System design defines the overall architecture, structure, and interaction between different components of the MeetSync application. This phase translates the system requirements into a blueprint that guides developers during implementation. The system design ensures efficiency, scalability, security, and ease of maintenance while providing a seamless user experience.

The system design ensures that:

- The software is modular and easy to maintain.
- Components are reusable and independently testable.
- The system supports scalability for future enhancements.
- UI/UX remains simple, intuitive, and responsive.

5.1 Primary Design Phase

The Primary Design Phase focuses on defining the overall architecture of the MeetSync system, identifying system users, major modules, and establishing the development blueprint required for implementation.

5.1.1 Identification of System Users

Two primary system users were identified:

1. User / Student

- Can register and log in
- Enter skills, interests, and education details

- Receive AI-based career recommendations
- View suggested skills, courses, and learning roadmap
- Track career progress through dashboard

2. Admin

- Manages career data, skills, and course resources
- Monitors user activity and system usage
- Maintains database integrity
- Controls system-level configurations

5.1.2 Identification of System Modules

Based on functional requirements, the following core modules were finalized:

- **User Authentication Module**
- **AI Career Recommendation Module**
- **Skill & Course Management Module**
- **Dashboard & Progress Tracking Module**
- **Admin Management Module**

Each module consists of related sub-features and backend API endpoints.

5.1.3 High-Level Architectural Planning

The system architecture was planned by identifying the interaction between frontend, backend,

AI logic, and database layers.

Key architectural considerations include:

- Frontend user interface built using **React.js**
- Secure routing and navigation using **React Router**

- Backend API development using **Node.js**
- AI logic for recommendation processing
- Database interaction using **PostgreSQL with Prisma ORM**

Overall architecture follows a layered approach:

UI Layer → API Layer → Business Logic → AI Engine → Database Layer

5.1.4 Database Schema Planning

Initial database schema planning was performed to ensure structured and scalable data storage:

- **Users Table** – Stores user profile, login credentials, skills, interests
- **Careers Table** – Stores career paths, descriptions, and requirements
- **Skills Table** – Contains skill lists mapped to career paths
- **Courses Table** – Learning resources and recommendations
- **UserProgress Table** – Tracks saved careers and learning progress

This design ensures data consistency and easy future expansion.

5.1.5 Wireframe & Screen Flow Planning

Before development, rough wireframes were created to understand user interaction flow:

- Home Page
- Login & Signup Pages
- Profile Setup (Skills & Interests)
- Career Recommendation Page
- Skill & Course Suggestions
- User Dashboard
- Admin Dashboard

This planning helped visualize navigation and user movement across the system.

5.2 Secondary Design Phase

The Secondary Design Phase focuses on transforming high-level design into detailed technical specifications including API structure, module interactions, and validation logic.

5.2.1 Detailed REST API Planning

RESTful APIs were designed for the following functionalities:

- User Registration & Login
 - Profile Creation and Update
 - Career Recommendation Generation
 - Skill & Course Retrieval
 - Dashboard Data Fetching
 - Admin Data Management

Each API follows:

- Input validation
 - Secure authentication checks
 - Role-based authorization
 - Structured JSON responses
 - Proper error handling

5.2.2 Business Logic Mapping

Core business logic includes:

- **AI-based Career Matching**

- Matches user skills and interests with career paths
- **Skill Recommendation Logic**
- Suggests beginner to advanced skill roadmap
- **Course Recommendation**
- Links relevant learning resources
- **Admin Privileges**
- Only admin can modify career, skill, and course data
- **User Restrictions**
- Users can view and update only their own profiles

5.2.3 Component-Based Design (React.js)

The frontend is divided into reusable and modular components:

- Navbar & Footer
 - Login / Signup Components
 - Profile Form
 - Career Recommendation Card
 - Skill Roadmap Component
 - Dashboard Widgets
 - Admin Panels

Advantages:

- Improved reusability
- Cleaner code structure
- Easy maintenance and scalability

5.2.4 Data Flow Design

The system follows the following data flow:

1. User submits input from frontend UI
2. Request is sent to backend API
3. Business logic processes the request
4. AI recommendation engine generates results
5. PostgreSQL database fetches or stores data
6. Frontend updates UI with results

5.2.5 Error & Exception Handling Design

The system handles common errors effectively:

- Invalid login credentials → Error message
 - Incomplete profile details → Validation alerts
 - No matching career found → Informative message
 - Unauthorized access → Access denied
 - Server failure → User-friendly error response

5.2.6 Security Design

Security measures implemented include:

- Password hashing using **Bcrypt**
 - Secure authentication using **JWT / Sessions**
 - Protected admin routes
 - Input sanitization to prevent attacks
 - Secure database access using Prisma ORM

5.3 User Interface Design (UI/UX)

The UI/UX design ensures simplicity, clarity, and a smooth user experience.

5.3.1 Design Principles Used

- **Simplicity** – Clean and minimal interface
 - **Consistency** – Uniform colors and layout
 - **Responsiveness** – Works on mobile, tablet, and desktop
 - **Accessibility** – Easy-to-read fonts and spacing
 - **User-Centered Design** – Simple navigation and clear flow

5.3.2 User-Side Screens Designed

1. Home Page – System overview
2. Signup Page – New user registration
3. Login Page – Secure authentication
4. Profile Setup Page – Skills & interests input
5. Career Recommendation Page
6. Skill & Course Suggestion Page
7. User Dashboard – Track saved careers and progress

5.3.3 Admin-Side Screens Designed

1. Admin Login Page
2. Admin Dashboard – System overview
3. Manage Careers Page

4. Manage Skills Page
5. Manage Courses Page
6. View User Data Page

5.3.4 UI Technology Stack

- **React.js** – Component-based frontend
 - **Tailwind CSS / CSS Modules** – Responsive styling
 - **React Router** – Page navigation
 - **Axios / Fetch API** – Backend communication

5.3.5 Overall UI Flow

- **User Flow:**

Home → Login → Profile Setup → Career Suggestions → Skills & Courses → Dashboard

- **Admin Flow:**

Login → Dashboard → Manage Careers → Manage Skills → Monitor Users

CHAPTER – 8

PROJECT CODES

Function.js

```
import { db } from "@lib/prisma";
import { ingest } from "./client";
import { GoogleGenerativeAI } from "@google/generative-ai";

const genAI = new GoogleGenerativeAI(process.env.GEMINI_API_KEY);
const model = genAI.getGenerativeModel({ model: "gemini-pro" });

export const generateIndustryInsights = ingest.createFunction(
  { name: "Generate Industry Insights" },
  { cron: "0 0 * * 0" }, // Run every Sunday at midnight
  async ({ event, step }) => {
    const industries = await step.run("Fetch industries", async () => {
      return await db.industryInsight.findMany({
        select: { industry: true },
      });
    });

    for (const { industry } of industries) {
      const prompt = `
        Analyze the current state of the ${industry} industry and provide insights in ONLY the following
        JSON format without any additional notes or explanations:
        {
          "salaryRanges": [
            { "role": "string", "min": number, "max": number, "median": number, "location": "string" }
          ],
          "growthRate": number,
          "demandLevel": "High" | "Medium" | "Low",
          "topSkills": ["skill1", "skill2"],
          "marketOutlook": "Positive" | "Neutral" | "Negative",
          "keyTrends": ["trend1", "trend2"],
          "recommendedSkills": ["skill1", "skill2"]
        }

        IMPORTANT: Return ONLY the JSON. No additional text, notes, or markdown formatting.
        Include at least 5 common roles for salary ranges.
        Growth rate should be a percentage.
        Include at least 5 skills and trends.
      `;
    }
  }
);
```

```

const res = await step.ai.wrap(
  "gemini",
  async (p) => {
    return await model.generateContent(p);
  },
  prompt
);

const text = res.response.candidates[0].content.parts[0].text || "";
const cleanedText = text.replace(/``(?:json)?\n?/g, "").trim();

const insights = JSON.parse(cleanedText);

await step.run(`Update ${industry} insights`, async () => {
  await db.industryInsight.update({
    where: { industry },
    data: {
      ...insights,
      lastUpdated: new Date(),
      nextUpdate: new Date(Date.now() + 7 * 24 * 60 * 60 * 1000),
    },
  });
});
}
}
);

```

CHAPTER – 9

PROJECT OUTCOMES

The successful implementation of the **MeetSync** will lead to the following outcomes:

9.1. User Module →

User Dashboard -

The User Dashboard provides an overview of the user's activities. It displays career recommendations, booked sessions, and progress insights. This helps users manage their career planning efficiently.



Fig. 9.1

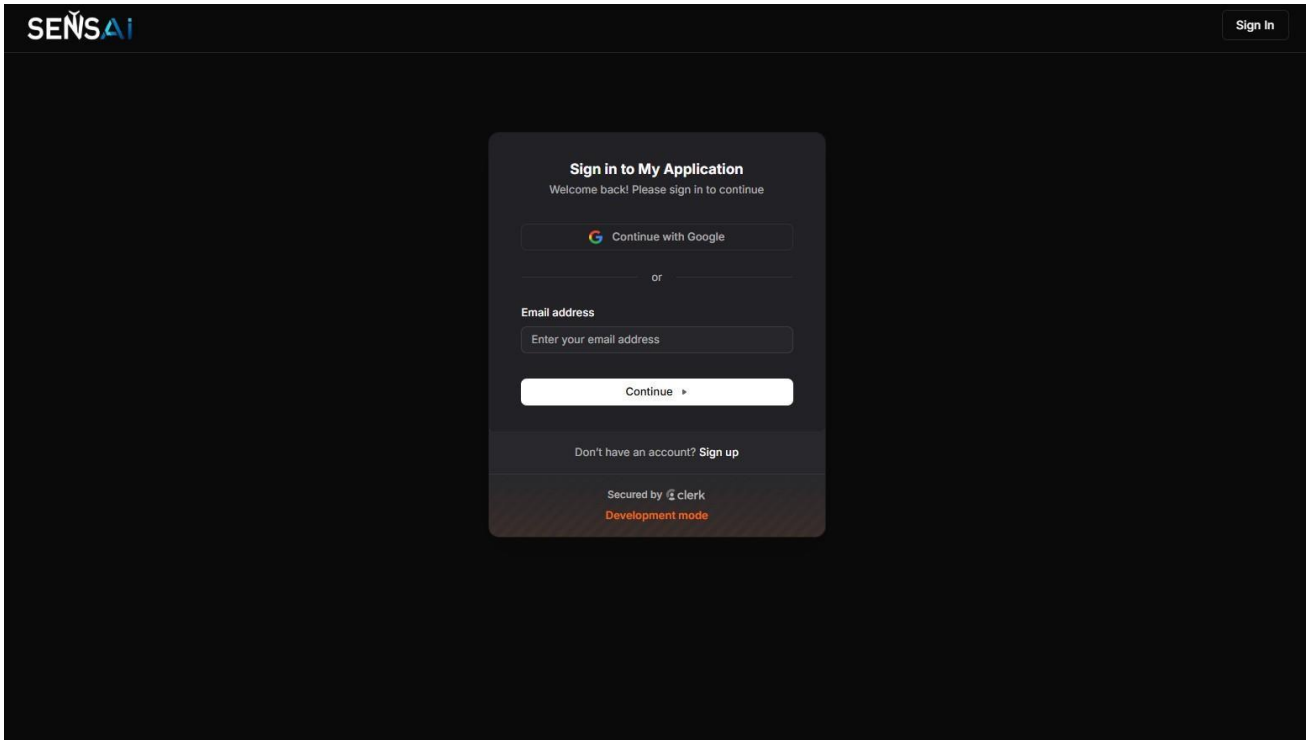


Fig. 9.2

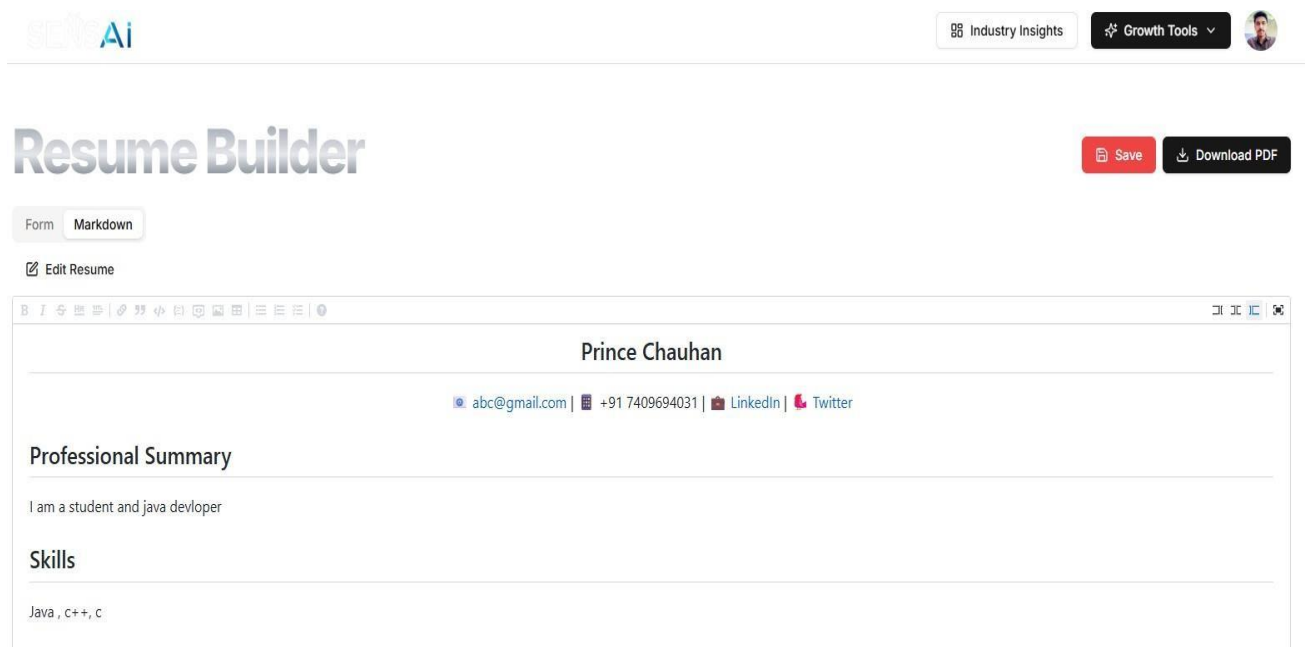


Fig. 9.3

seerAi

Industry InsightsGrowth Tools

Back to Cover Letters

Create Cover Letter

Generate a tailored cover letter for your job application

Job Details

Provide information about the position you're applying for

Company Name

Enter company name

Job Title

Enter job title

Job Description

Paste the job description here

Generate Cover Letter

Fig. 9.4



Fig.9.5

➤ Enhanced User Experience

The MeetSync platform is designed to provide a smooth and user-friendly experience for users. The system offers easy navigation, clear dashboards, and quick access to career guidance features. Personalized recommendations and organized modules help users interact with the platform efficiently and confidently. (Fig. 9.6).

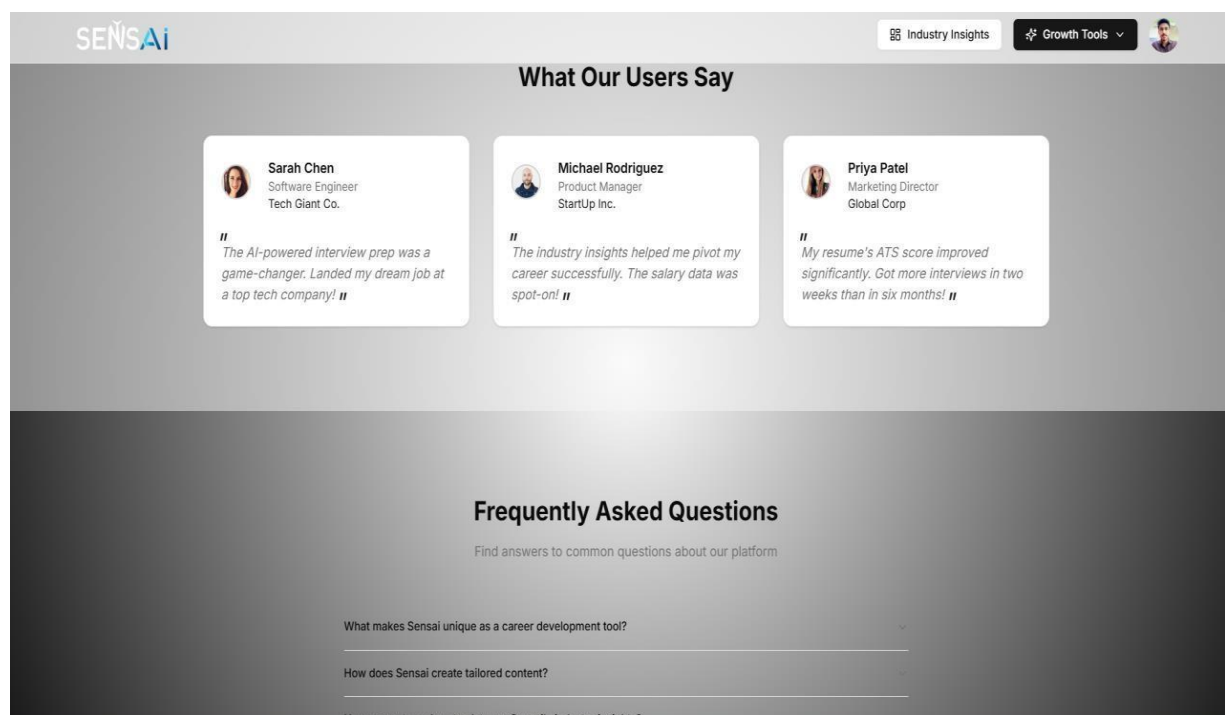


Fig. 9.6

➤ Progress Tracking & Analytics

This feature helps users monitor their career development over time. It provides insights into performance, completed activities, and improvement areas using analytics. Users can track their progress and make informed decisions to enhance their career growth. (fig. 9.7).

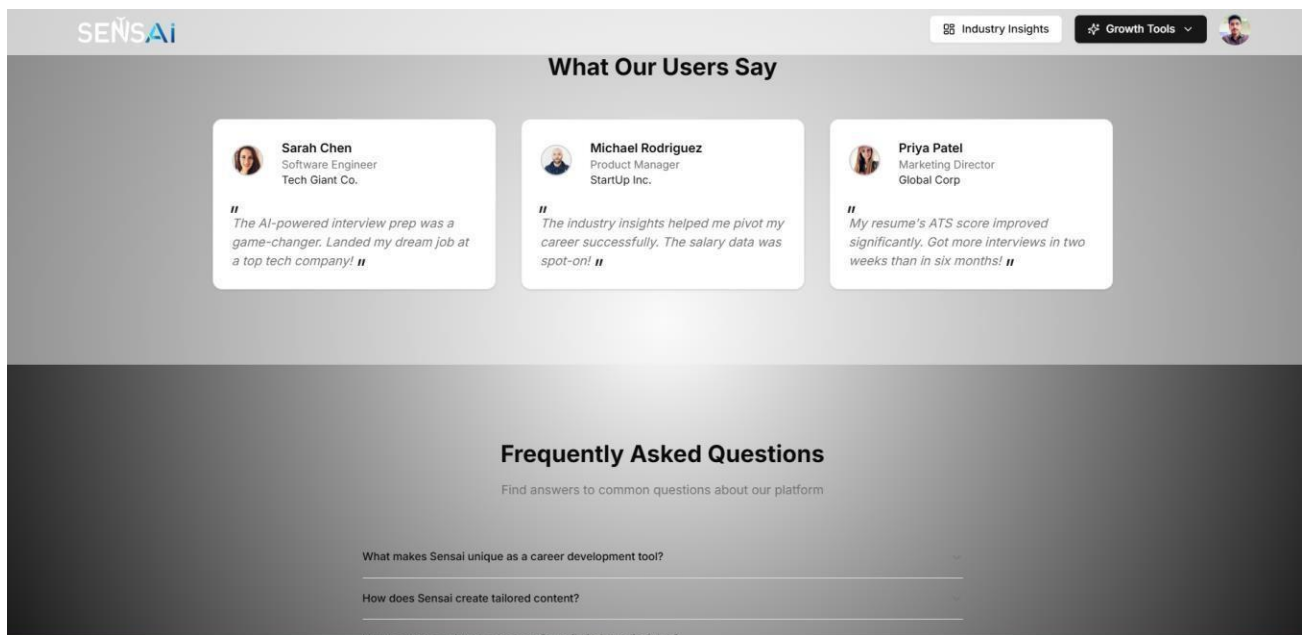


Fig. 9.7

➤ Performance Monitoring

This feature continuously monitors the user's activities on the MeetSync platform. It tracks completed career guidance sessions, mock interview performance, resume updates, and skill assessment results. By analyzing user interactions, the system identifies strengths and areas that require improvement. This helps users clearly understand how their skills are developing over time. Performance monitoring enables users to stay motivated and focused on their career goals through structured feedback

➤ Progress Reports

The Progress Reports section provides detailed insights into the user’s career growth journey. It generates reports based on learning progress, interview practice scores, and completed milestones. These reports allow users to evaluate their improvement and identify gaps in skills or preparation. With clear visual and data-driven summaries, users can plan future actions more effectively. This feature ensures continuous improvement through data-based career planning.

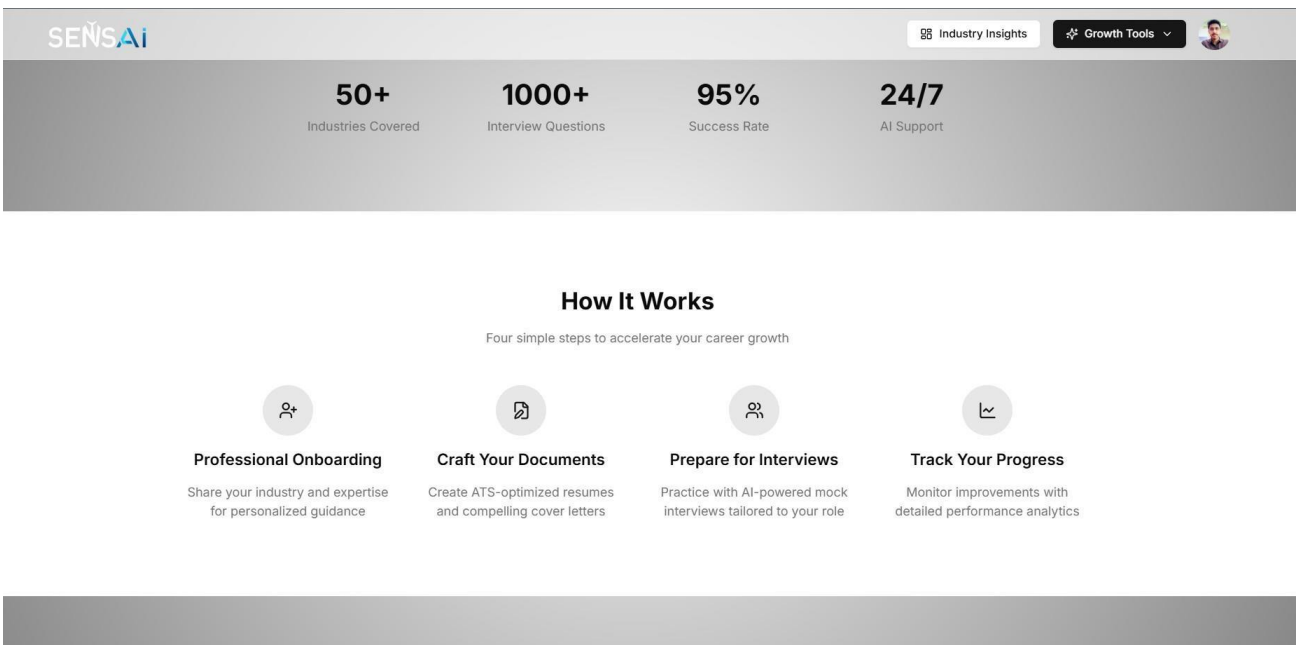


Fig. 9.8

9.2 Admin Module → Dashboard, Manage Account etc.

➤ Admin Dashboard

The Admin Dashboard provides a centralized view of the entire MeetSync system. It displays key information such as total users, active sessions, system usage, and performance statistics. This helps administrators monitor platform activity efficiently and make informed management decisions.

➤ User Management

This feature allows the admin to manage registered users on the platform. Administrators can view user details, update user status, or remove inactive or unauthorized accounts. User management ensures data accuracy, security, and smooth platform operation.

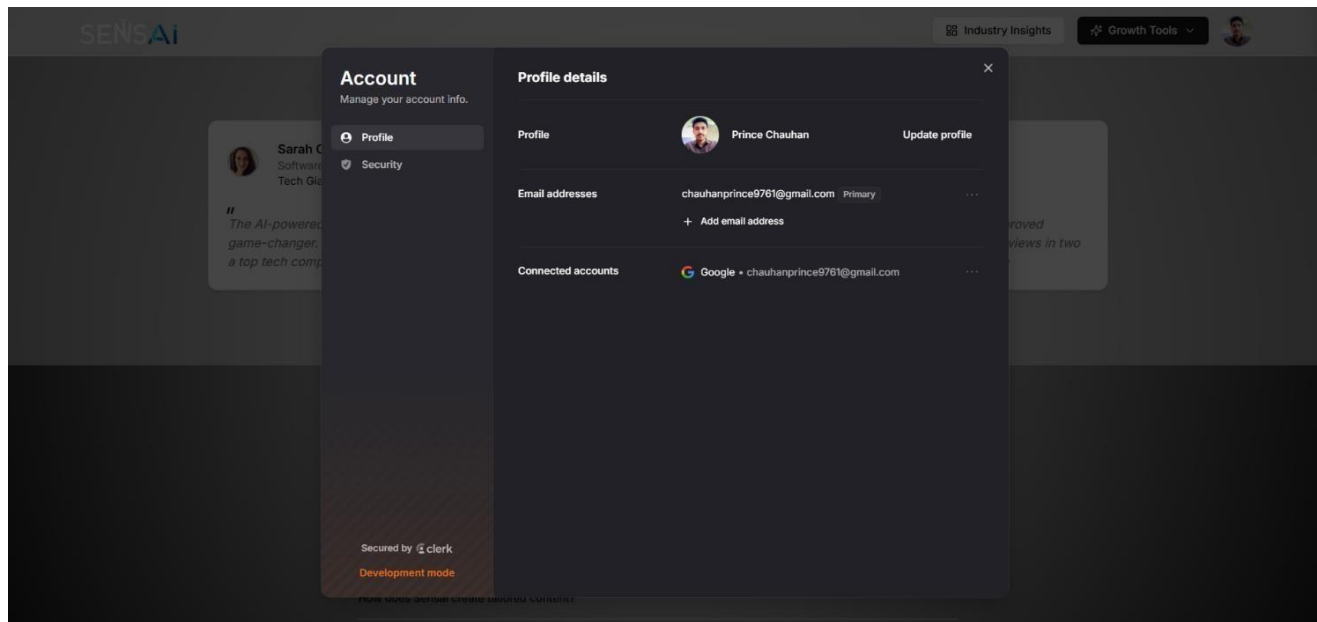


Fig. 9.9

9.3 Database Module

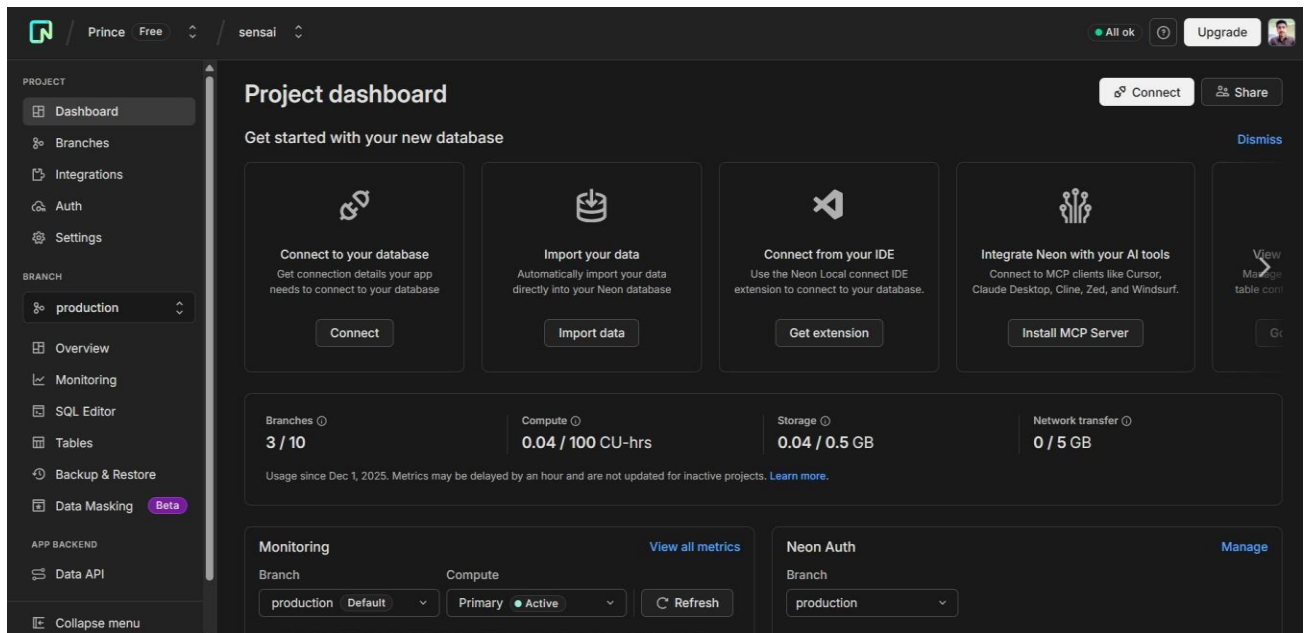


Fig. 9.10

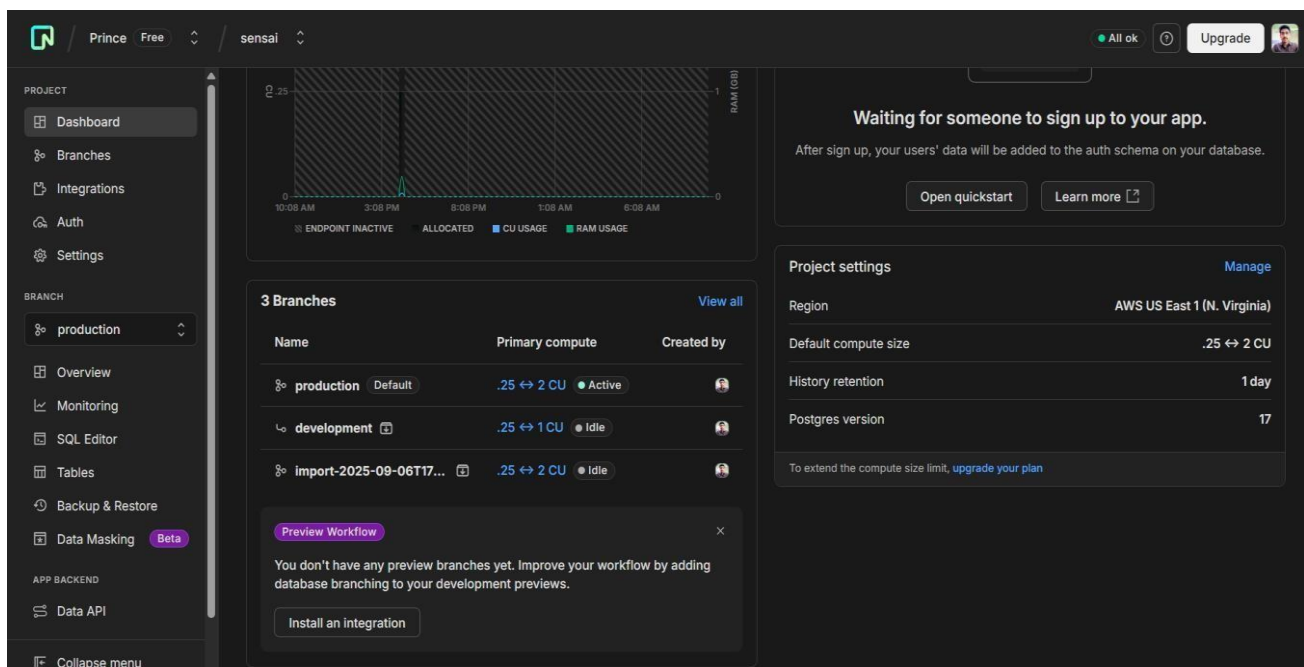


Fig. 9.11

Prince

Free

All ok

Upgrade

ORGANIZATION

Projects

People

Billing

Integrations

Settings

Feedback

Collapse menu

Prince's projects

New project

Import data

Compute 0.04 CU-hrs

Storage 0.07 GB

History 0 GB

Network transfer 0 GB

Usage since Dec 1, 2025. Metrics may be delayed by an hour and are not updated for inactive projects. [Learn more.](#)

Q Search...

Name	Region	Created at	Storage	Compute last active at	Branches	Integrations
sensai	AWS US East 1 (N. Virginia)	Aug 25, 2025 9:25 pm	38.9 MB	Dec 29, 2025 4:18 pm	3	Add

Passwordless connect with psql:

\$ psql -h pg.neon.tech

Fig. 9.12

46

CHAPTER – 10

TESTING OF MEETSYNCPROJECT

Software testing is one of the most critical phases in the software development life cycle. It ensures that the system performs accurately, provides reliable AI-based recommendations, and functions smoothly across different scenarios. The **MeetSync System** was thoroughly tested using **Manual Testing**, focusing on validating functional correctness, AI recommendation flow, user experience, and database accuracy.

10.1 Introduction to Testing

Testing is a systematic process carried out to verify that the software application meets all specified requirements and performs reliably under real-world conditions. It helps identify defects, incorrect logic, data inconsistencies, usability issues, and security flaws before deployment.

For the **MeetSync Application**, testing was performed with the following objectives:

- To ensure correct user registration, login, and profile creation
- To validate AI-based career recommendation accuracy
- To verify skill and course suggestion functionality
- To confirm admin operations such as managing careers, skills, and courses
- To ensure secure data storage in PostgreSQL database
- To check responsiveness, usability, and browser compatibility

The primary goal of testing was to deliver a **stable, intelligent, and user-friendly career guidance platform**.

10.2 Types of Testing Conducted

Multiple testing techniques were applied to validate all system modules.

1. Functional Testing

Ensures that every feature of the MeetSync works as per requirement specifications.

Key areas tested:

- User registration and authentication
- Google & Email login
- Profile creation (skills, interests, education)
- AI career recommendation generation
- Skill roadmap display
- Course suggestions
- Dashboard data display
- Admin operations (Add/Edit/Delete careers, skills, courses)

2. Usability Testing

Focused on user interaction, ease of use, and navigation clarity.

Observed tasks:

- Ease of completing profile setup
- Clarity of career recommendations
- Navigation between dashboard sections
- Readability of text and instructions
- Responsiveness on mobile and desktop devices

3. Validation Testing

Ensures the system accepts valid input and rejects invalid data.

Examples:

- Preventing duplicate email registrations
- Blocking login with incorrect credentials
- Mandatory skill and interest selection
- Validating profile completeness before recommendations
- Admin-only access restrictions

4. Compatibility Testing

Checks application behavior across different browsers.

Browsers tested:

- Google Chrome
- Mozilla Firefox

UI consistency was verified for:

- Layout
- Fonts and colors
- Button alignment
- Navigation behavior

5. Database Testing

Ensures accurate CRUD operations in **PostgreSQL database**.

Database testing included:

- Storing user profiles correctly
- Saving AI-generated recommendations
- Updating skill progress
- Fetching dashboard data
- Admin data modifications

Database accuracy is critical as AI recommendations depend on stored user data.

10.3 Test Plan and Execution

The test plan defines testing scope, strategy, environment, and execution approach.

Testing Objectives

- Validate user and admin functionalities
- Ensure AI recommendation accuracy
- Verify data integrity and API responses
- Identify UI and usability issues

Testing Strategy

- Manual testing of all modules
- Use of valid, invalid, and edge-case inputs
- Browser testing on Chrome and Firefox

- Database verification using PostgreSQL tools
- Backend log monitoring for API validation

Test Environment

- Frontend: React.js
- Backend: Node.js
- Database: PostgreSQL
- Authentication: NextAuth / JWT
- OS Used: Windows 10
- Browsers: Chrome & Firefox

Execution Approach

Testing was conducted in phases:

Phase 1: Module Testing

Each module (Authentication, AI Recommendation, Dashboard, Admin) was tested independently.

Phase 2: Integration Testing

Combined workflows tested:

- Login → Profile Setup → Career Recommendation → Dashboard

Phase 3: System Testing

End-to-end system testing performed.

Phase 4: Regression Testing

Modules re-tested after bug fixes.

10.4 Results and Reports – Manual Test Cases (15 Test Cases)

Below are the manual test cases covering the entire system.

TABLE: Manual Test Cases

Test Case ID	Test Scenario	Input	Expected Result	Actual Result	Status
TC01	Register new user	Valid user details	User registered successfully	As expected	PASS
TC02	Register with existing	Duplicate email	Error message	As expected	PASS

Test Case ID	Test Scenario	Input	Expected Result	Actual Result	Status
	email		shown		
TC03	Login with correct credentials	Valid email & password	Redirect to dashboard	Works correctly	PASS
TC04	Login with wrong password	Invalid password	Error displayed	Works correctly	PASS
TC05	Google OAuth login	Google account	Successful login	Works correctly	PASS
TC06	Profile creation	Skills & interests	Profile saved	Works correctly	PASS
TC07	Generate career recommendation	Completed profile	Career suggestions shown	Correct output	PASS
TC08	View skill roadmap	Click career	Skill roadmap displayed	Works correctly	PASS
TC09	View course suggestions	Select skill	Courses displayed	Works correctly	PASS
TC10	Dashboard load	User login	Data loaded	Works correctly	PASS
TC11	Update user profile	Edit skills	Profile updated	Updated correctly	PASS
TC12	Admin login	Admin credentials	Admin dashboard access	Works correctly	PASS
TC13	Add career (Admin)	Career details	Career added	Works correctly	PASS
TC14	Edit skill (Admin)	Updated skill info	Skill updated	Works correctly	PASS
TC15	Validate database entry	User & career data	Saved in PostgreSQL	Correctly saved	PASS

CHAPTER – 11

PROPOSED TIME DURATION

PROPOSED TIME DURATION

The development of the **MeetSync Application** is planned to be completed within **8–9 weeks**, following a well-structured and systematic timeline. The tentative phase-wise duration is outlined below:

1. Phase 1 – Requirement Analysis & Planning (2 weeks)

- Understanding the project objectives and problem statement related to career guidance.
- Gathering functional and non-functional requirements from users.
- Finalizing features such as AI-based career recommendations, skill roadmap, dashboard, and authentication.
- Selecting the technology stack (React, Node.js, PostgreSQL, AI logic).
- Preparing project roadmap and assigning responsibilities among team members.

2. Phase 2 – Frontend Development (2 weeks)

- Designing and developing user-friendly UI using **React.js**.
- Implementing pages such as Home, Login, Signup, Profile Setup, Career Recommendation, and Dashboard.
- Creating reusable components for skills, courses, and recommendations.
- Ensuring responsive design for desktop, tablet, and mobile devices.

3. Phase 3 – Backend Development (3 weeks)

- Setting up backend server using **Node.js**.
- Developing RESTful APIs for authentication, profile management, and career recommendations.
- Implementing AI logic for analyzing user skills and interests.
- Handling role-based access for users and admin.
- Implementing security features such as authentication, authorization, and error

handling.

4. Phase 4 – Database Integration (2 weeks)

- Designing relational database schema using **PostgreSQL**.
- Integrating database using **Prisma ORM**.
- Storing user profiles, career paths, skills, and course data.
- Managing dashboard data and user progress tracking.
- Optimizing queries for efficient data retrieval.

5. Phase 5 – Testing & Debugging (2 weeks)

- Performing manual testing of all modules.
- Conducting functional, validation, and usability testing.
- Verifying AI recommendation accuracy and dashboard updates.
- Fixing bugs and improving system performance.
- Ensuring browser compatibility and responsive behavior.

6. Phase 6 – Deployment, Maintenance & Updates (1 week, ongoing)

- Deploying frontend and backend on cloud platforms (e.g., Vercel).
- Monitoring system performance and user activity.
- Fixing post-deployment issues.
- Preparing for future enhancements such as advanced AI models, career analytics, and personalized learning paths.

CONCLUSION

The MeetSync project successfully demonstrates the application of modern web technologies and artificial intelligence concepts to address the growing challenge of career decision-making. The system provides a structured, intelligent, and user-friendly platform that assists individuals in identifying suitable career paths based on their skills, interests, and educational background. By automating career guidance, the project reduces dependency on traditional counseling methods, making career support more accessible and scalable.

The application integrates a responsive frontend built using React.js with a secure and efficient backend developed using Node.js and PostgreSQL. The use of AI-driven recommendation logic enables personalized career suggestions, skill roadmaps, and relevant learning resources, ensuring that users receive meaningful and actionable guidance. Secure authentication, proper data validation, and structured database management enhance the reliability and safety of the system.

Throughout the development process, emphasis was placed on usability, performance, and data integrity. Comprehensive manual testing confirmed that all core functionalities, including user authentication, profile management, career recommendations, and admin operations, perform as expected. The modular and component-based architecture ensures maintainability and allows easy future enhancements.

Overall, the MeetSync project fulfills its objectives by delivering a robust, efficient, and intelligent career guidance system. It serves as a practical example of how artificial intelligence and full-stack development can be combined to solve real-world problems.

REFERENCES

Books

1. Sommerville, Ian. *Software Engineering*, 10th Edition, Pearson Education.
2. Pressman, Roger S. *Software Engineering: A Practitioner's Approach*, McGraw-Hill.
3. Stevens, W., Myers, G., & Constantine, L. *Structured Design: Fundamentals of a Discipline of Computer Program and Systems Design*.
4. Date, C. J. *An Introduction to Database Systems*, Pearson.
5. Elmasri, Ramez & Navathe, Shamkant. *Fundamentals of Database Systems*, Pearson.
6. Martin Fowler. *Patterns of Enterprise Application Architecture*.
7. Mark Richards. *Software Architecture Patterns*.
8. Grady Booch. *Object-Oriented Analysis and Design With Applications*.
9. Kyle Simpson. *You Don't Know JS (Series)*, O'Reilly Media.
10. Ethan Brown. *Learning JavaScript*, O'Reilly Media.

Websites

11. React Official Documentation – <https://react.dev>
12. Node.js Official Documentation – <https://nodejs.org>
13. Express.js Guide – <https://expressjs.com>
14. MongoDB Atlas Documentation – <https://www.mongodb.com/docs>
15. JavaScript MDN Web Docs – <https://developer.mozilla.org>
16. W3Schools Web Technology Tutorials – <https://www.w3schools.com>
17. GeeksforGeeks Web Development Notes – <https://www.geeksforgeeks.org>
18. Tutorialspoint MERN Concepts – <https://www.tutorialspoint.com>
19. FreeCodeCamp Full-Stack Articles – <https://www.freecodecamp.org>
20. DigitalOcean Tutorials – <https://www.digitalocean.com/community/tutorials>
21. Stack Overflow Developer Community – <https://stackoverflow.com>

22. REST API Design Best Practices – <https://restfulapi.net>
23. OWASP Web Security Guidelines – <https://owasp.org>
24. Google Developers Web Fundamentals – <https://developers.google.com/web>
25. CSS Tricks Flexbox & Grid Guide – <https://css-tricks.com>
26. Medium Web Development Articles – <https://medium.com>
27. GitHub Documentation – <https://docs.github.com>
28. Postman API Testing Docs – <https://learning.postman.com>

Research & Technical Articles

29. ACM Digital Library – Cloud Database Access Performance Studies.
30. IEEE Papers on Full-Stack Web Application Architecture.
31. Journal of Web Engineering – UI/UX Usability Research.
32. ResearchGate Articles on Booking Systems and Reservation Software Models.
33. SpringerLink – Agile Development Methodologies.
34. Traversy Media – MERN Stack & REST API Tutorials.
35. Net Ninja – React.js Tutorials.
36. Programming with Mosh – Node.js & MongoDB videos.
37. Academind – React & Backend API Development.
38. CodeWithHarry – Web Development Fundamentals.

YouTube Tutorials

39. Hitesh Choudhary – Full Stack Development Concepts.
40. Enterprise Car Rental Booking Flow (enterprise.com).
41. ZoomCar Booking Process UI Study (zoomcar.com).
42. Ola Drive Self-Car Rental Flow (olaelectric.com).
43. Hertz Car Rental Business Model Reference (hertz.com).

Additional Online Sources

44. UI Patterns & Design Principles – <https://uipatterns.com>
45. Dev.to Full Stack Development Articles – <https://dev.to>

